

AI/ML INTERN — DAY 3 TASKS

Task 1 — Human Pose Detection

Objective:

Build a human pose detection system using MediaPipe and OpenCV to detect body landmarks and display skeleton tracking in real time.

Tools Used

- Python
- MediaPipe
- OpenCV
- VS Code

Features Implemented

- Detected human body landmarks using MediaPipe Pose.
- Displayed skeleton tracking with connected body points.
- Processed live camera feed using OpenCV.
- Visualized pose estimation successfully.

Output Screenshot

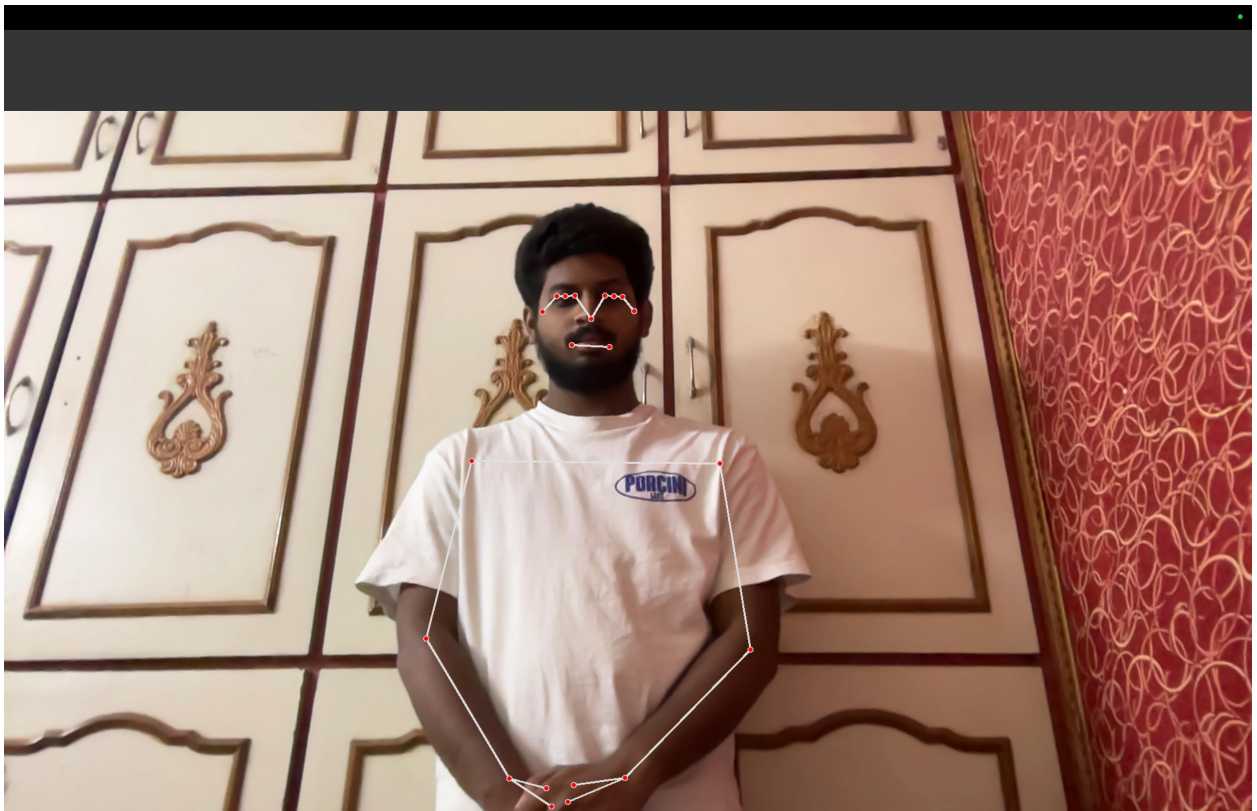


Figure: Human pose detection and skeleton tracking output.

Conclusion

The human pose detection system was successfully implemented using MediaPipe and OpenCV. The system accurately detects body landmarks and displays skeleton tracking in real time.

Task 2 — Body Measurement Logic

Objective

Understand how AI systems calculate human body measurements using pose detection landmarks from MediaPipe.

1. Shoulder Width Calculation

Shoulder width is calculated using the distance between the left shoulder and right shoulder landmarks.

Formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Python Example

```
import math
```

```
x1, y1 = left_shoulder.x, left_shoulder.y
```

```
x2, y2 = right_shoulder.x, right_shoulder.y
```

```
shoulder_width = math.sqrt((x2 - x1)**2 + (y2 - y1)**2)
```

```
print("Shoulder Width:", shoulder_width)
```

2. Height Estimation

Height estimation uses the distance between the head landmark and foot landmarks.

3. Waist Estimation

Waist size is estimated using the distance between the left hip and right hip landmarks.

Applications

- Virtual fitting rooms
- Fitness tracking
- Smart tailoring
- Fashion recommendation systems

Task 3 — AI Research

1. Virtual Try-On AI Systems

Virtual try-on AI systems allow users to upload their image and digitally try clothes using AI-based body tracking and clothing mapping.

How Virtual Try-On Works

1. Body Detection using pose estimation
2. Clothing Segmentation
3. Cloth Mapping onto the body
4. Real-time visualization

Technologies Used

- MediaPipe
- OpenCV
- Deep Learning
- CNN Models
- GAN Models

2. Fashion Recommendation Model

Fashion recommendation systems suggest clothes based on user preferences, body type, and fashion trends.

How Recommendation Models Work

- User uploads image
- AI detects body measurements
- AI predicts clothing size
- Recommendation engine suggests outfits

Advantages

- Personalized recommendations
- Better shopping experience
- Reduced clothing returns
- Faster outfit selection